Venkatesh Akula

Fourth Year Undergraduate
Department of Computer Science and Engineering, IIT Kanpur

akulav22@iitk.ac.in | venkateshakula1729
+91-9014764100 | Venkatesh Akula

Program	Institute	Year	CPI/%
B.Tech CSE	Indian Institute of Technology Kanpur	2022–Present	**9.34**/10
TSBIE (XII)	Narayana Junior College, Hyderabad	2022	**98.8%**
TSBSE (X)	Narayana School, Hyderabad	2020	**10**/10

Scholastic Achievements

- Conferred with **Academic Excellence Award** twice by IIT Kanpur for outstanding academic performance 2022–2024
- Secured **All India Rank 477** in **JEE Advanced 2022**
- Qualified for **Indian National Mathematics Olympiad (INMO)**, by excelling in **RMO 2019** conducted by HBCSE
- Qualified for **INAO 2020** (HBCSE), top **250** across India
- Qualified for **INChO 2021** (HBCSE), top **1%** in the Nation
- **Awarded KVPY SA 2020** fellowship, with **AIR 148**
- Secured **All India Rank of 769** in **JEE Mains 2022**
- **Awarded KVPY SX 2021** fellowship, with **AIR 552**
- **State 1st Rank** in **(AIMED) Maths Scholarship Test**
- Recipient of Reliance Scholarship for being in **National Top 1%**

Work Experience

Adobe ML Systems Research *May'25 - Jul'25*
Intern Research Assistant, Big Data Experience Lab

- **Optimized multi-agent LLM inference workflows** to improve **inter-agent communication efficiency**, **context handling**, **coordination**, and overall **inference quality**
- Identified systemic **latency and communication bottlenecks** in **multi-agent LLM systems** through controlled experiments
- Evaluated agent communication frameworks including **AutoGen**, **LangChain**, and prompt optimization frameworks such as **DSPy**, **TextGrad**, and **AdalFlow**, analyzing **response quality**, **token usage**, and **communication overhead**
- Implemented a novel **token-efficient, context-aware training and inference pipeline** that preserved accuracy while reducing **compute cost** and **end-to-end latency**
- Achieved $\approx$ **35%** reduction in **decode tokens** with $\approx$ **1.5**× **latency** improvement across **QA**, **debate**, and **code-generation** pipelines while maintaining **accuracy**; under **multi-turn** and **long-context inference** settings, outperforming **state-of-the-art (SOTA) baselines**

Programming Achievements

- Ranked **1st** in **IIT Kanpur** in Advent of Code 2024 among **200+** participants by solving all 50 problems in C++/Python
- Reached a maximum **Codeforces** rating of **1729** with rank **233** (CF-1007 Div 2) and **268** (CF-175 Div 2) on Progmatician
- Reached a maximum **CodeChef** rating of **1821** with Global rank **92** (Starters 179) and **102** (Starters 177) on venkey1729

Positions Of Responsibility

Executive, Stamatics, IIT Kanpur *Nov'23 - Jul'24*

- Led **Mathemania'23** problem-setting & operations (**500+** participants); organized **research talks** on applied mathematics

Secretary | Career Development Wing *Jun'23 - Apr'24*

- Organized **tests**, **workshops** and **guidance sessions** for UG students to help them prepare for their **career** paths

Student Guide | Counselling Service *Aug'23 - Jul'24*

- Mentored a group of **6 freshmen** academic planning and social onboarding to facilitate a smooth transition into campus life

Selected Projects

Stock Forecasting with Deep Learning *Mar'24 – Apr'24*
Programming Club, IIT Kanpur

- Built a prediction model using **LSTM** (**RNN**) to capture complex **temporal dependencies** in stock data
- Applied **data preprocessing** including **normalization**, **scaling**, **outlier clipping**, and missing-value handling to optimize training
- Trained on **10 years** of data to predict the **closing stock price** for the next **2 days**, given the past **50 days** as input
- Achieved **6.5% RMSE** and **95.3% directional accuracy**

Deep Fake Detection *Jan'25 – Apr'25*
Course Project, CS776

- Built and trained multiple **CNN baselines** (VGG19, ResNet50, MobileNetV1, EfficientNet-B0) for robust **deepfake detection** on CIFAKE/DFRI, achieving up to **94%** clean accuracy
- Implemented **adversarial attacks** (FGSM, PGD, One-Pixel) in **PyTorch** to evaluate robustness of **CNN** detectors
- Designed a **DeiT Vision Transformer** and a hybrid **CNN + Transformer** model achieving **98%** clean and **75%** robust accuracy under **adversarial attacks**

Building GemOS *Aug'24 – Nov'24*
Course Project, CS330

- Built a **grep-like tool**, **syscall tracer**, and **process-chaining shell** using **strace**, **file I/O**, **fork/exec**, and **pipes**
- Implemented custom **malloc()** and **free()** for dynamic memory management, handling fragmentation via linked lists
- Implemented **mremap()** syscalls by allocating pages, updating page tables, and performing **multi-level page table walks**
- Developed a **round-robin scheduler** and full semaphore functionality via **sem_wait()** and **sem_post()**

High Performance Computing *May'25 – Jun'25*
Course Project, CS633

- Developed an MPI-based code to compute extrema over 3D **time-series datasets** using **3D domain decomposition**
- Implemented custom MPI subarray datatypes for fast **parallel I/O**, replacing a **sequential I/O** bottleneck and achieving **8**× faster data loading (up to $64 \times 64 \times 64$)
- Achieved $\approx$ **3.4**× speedup on 16 processes with optimal scaling up to 16 cores using **parallelization techniques**

Computer Networks *Jan'25 – Apr'25*
Course Project, CS425

- Developed a multi-threaded **TCP chat server** in C++ with user **auth**, **/broadcast**, **/msg**, and **mutexes** for concurrency
- Built **iterative/recursive DNS resolvers** and performed root → TLD → authoritative queries for **domain-to-IP mapping**
- Built a **raw-socket TCP client** in C++ to perform the 3-way handshake handling **SYN, SYN-ACK, ACK** at packet level

Technical Skills

- **Programming Languages:** C/C++, Python, JavaScript, Solidity, HTML, CSS, Verilog
- **Software and Libraries:** React, Node.js, Express.js, MongoDB, Matplotlib, Pandas, sklearn, PyTorch

Relevant Courses

*: Excellent Performance | †: Ongoing

Data Structures and Algorithms	Parallel Computing *	Operating Systems	Computer Organisation
Probability & Statistics	Mathematical Finance *	Deep Learning for CV	Algo Information Theory *
Intro to ML *	Linear Algebra & ODE *	Computer Networks *	Financial Economics †
Logic in Computer Science	Macroeconomics †	Software Devops	Discrete Mathematics